

David Elijah Umaru

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EDUCATION

Colorado State University, Fort Collins, CO

Master of Science in Computer Engineering

May 2028 (expected)

Bachelor of Science in Computer Engineering, Minor in Computer Science GPA: 3.92/4.0

May 2026

- Concentration in Embedded Systems; Dean's List (6 semesters).

RESEARCH

Heavy Vehicle Cybersecurity Lab, Colorado State University

Aug 2026 – Present

Graduate Researcher, Advisor: Dr. Jeremy Daily

- Conducting graduate research on the cybersecurity of DC fast-charging systems for electric vehicles.
- Practiced CAN message injection and introductory fuzzing on a vehicle network testbed to study how control units respond to unexpected or malicious traffic.
- Participated in the CyberAuto Challenge and CyberTractor, hands-on vehicle cybersecurity training events.

EXPERIENCE

Department of Electrical & Computer Engineering, Colorado State University

Aug 2026 – Present

Graduate Teaching Assistant, ECE 205 Analog Circuit Design

- Lead weekly labs for about 60 students, teaching hands-on use of oscilloscopes, function generators and multimeters.
- Troubleshoot student circuits at the bench, coaching students to isolate wiring and component faults step by step.
- Grade lab reports and give written feedback on measurement accuracy and comparison to theoretical predictions.

Data Science Research Institute, Colorado State University

June 2025 – Aug 2026

HPC Developer Intern

- Diagnosed and resolved Linux system and job-scheduling issues for researchers on the university's computing cluster.
- Verified server configurations and ran operational tests while commissioning new hardware during an HPC cluster rebuild.
- Wrote user guides and troubleshooting docs (GitHub, Read the Docs), improving support efficiency by about 50%.

PROJECTS

Formula SAE Ram Racing, Electric Vehicle Team

Sept 2024 – Present

Electrical Team Member; Low Voltage Electrical Lead, Sept 2024 – May 2026

- Designed the 12V low-voltage electrical system and a 500 kbps CAN network with a custom message database, connecting the vehicle control unit, motor controller and battery management system.
- Designed the fault-latch PCB and its latching logic, then integrated and tested it with the insulation monitoring device and brake plausibility device in the shutdown circuit.
- Built a benchtop powertrain test stand with E-stop and precharge circuit to validate motor-controller startup sequencing; designed a power distribution PCB in KiCad.
- Developed vehicle control logic in MATLAB/Simulink for the motor controller; secured a New Eagle hardware sponsorship and currently mentor two new electrical members.

RamBots, Senior Design Capstone

Aug 2025 – May 2026

Embedded Systems Developer and Safety Officer

- Designed a PCB for power distribution and signal routing to the joint motors of a 5-DOF robotic arm built by a 6-member team.
- Developed real-time C++ control firmware for inverse kinematics, joint coordination and gripper control, plus a web interface for remote monitoring.
- Integrated Teensy 4.1, ODrive controllers and Raspberry Pi 4 for the quadruped platform, debugging SPI/UART links; served as team Safety Officer, writing safety procedures for the 13-person team.

SKILLS

Embedded

C/C++ for microcontrollers, Teensy 4.1, ESP32, Raspberry Pi, PCB design (KiCad), wiring harnesses

Protocols

CAN, J1939, SPI, I2C, UART, Ethernet

Security

CAN message injection, introductory CAN fuzzing, ISO/SAE 21434 concepts

Tools

Python, MATLAB/Simulink, Linux, Git/GitHub, Quartus Prime, Cadence Virtuoso

HONORS AND ACTIVITIES

Tesla Battery Sponsorship (2025) | Fry Family EE Scholarship | Carl Wilsen ECE Scholarship | NSBE | ColorStack